

US Stock Quant Toolkit: Quantitative Strategy Backtesting Project

Executive Summary

This is a Python-based quantitative research project that downloads real market data, tests multiple rule-based strategies, and compares return and risk profiles.

Data Source

The project uses yfinance to download historical price data for SPY or other US-listed stocks and ETFs.

Data Range

Full period: 2018-01-02 to 2026-05-27.

Train period: 2018-01-01 to 2021-12-31.

Test period: 2022-01-01 to 2026-05-27.

Strategy Design

The tested strategies are Buy and Hold, 20/60 Moving Average, 50/200 Moving Average, RSI Mean Reversion, and 6-Month Momentum.

Backtesting Methodology

The backtest uses daily returns. All signals are shifted by one day using shift(1) to avoid look-ahead bias. The transaction cost assumption is transaction_cost = 0.001. The sample is split into training and testing periods.

Performance Metrics

The report includes cumulative return, annualized return, annualized volatility, Sharpe Ratio, maximum drawdown, Calmar Ratio, win rate, number of trades, and turnover.

Empirical Results: Full Sample

Strategy	Cumulative Return	Annualized Return	Sharpe Ratio	Maximum Drawdown	Calmar Ratio
20/60 Moving Average	87.75%	7.81%	0.64	-27.12%	28.80%
50/200 Moving Average	103.14%	8.83%	0.57	-33.72%	26.18%
6-Month Momentum	72.28%	6.71%	0.54	-23.96%	28.00%
Buy and Hold	216.92%	14.76%	0.77	-33.72%	43.78%
RSI Mean Reversion	65.74%	6.22%	0.40	-28.34%	21.94%

Empirical Results: Train/Test

Strategy	Cumulative Return	Annualized Return	Sharpe Ratio	Maximum Drawdown	Calmar Ratio
20/60 Moving Average	87.75%	7.81%	0.64	-27.12%	28.80%
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Buy and Hold	216.92%	14.76%	0.77	-33.72%	43.78%
RSI Mean Reversion	65.74%	6.22%	0.40	-28.34%	21.94%
20/60 Moving Average	65.43%	13.41%	1.04	-12.44%	107.82%
50/200 Moving Average	34.90%	7.77%	0.44	-33.72%	23.05%
6-Month Momentum	31.33%	7.05%	0.53	-23.96%	29.43%
Buy and Hold	89.21%	17.28%	0.83	-33.72%	51.26%
RSI Mean Reversion	32.31%	7.25%	0.43	-28.34%	25.58%
20/60 Moving Average	13.49%	2.93%	0.26	-26.42%	11.10%
50/200 Moving Average	50.59%	9.81%	0.72	-18.76%	52.28%
6-Month Momentum	31.18%	6.40%	0.56	-22.84%	28.01%
Buy and Hold	67.49%	12.51%	0.71	-24.50%	51.05%
RSI Mean Reversion	25.27%	5.28%	0.37	-19.29%	27.38%

Key Findings

Highest cumulative return: Buy and Hold (216.92%).

Lowest drawdown: 6-Month Momentum (-23.96%).

Highest Sharpe Ratio: Buy and Hold (0.77).

Best testing-period Sharpe Ratio: 50/200 Moving Average (0.72).

Transaction costs reduce the performance of high-turnover strategies.

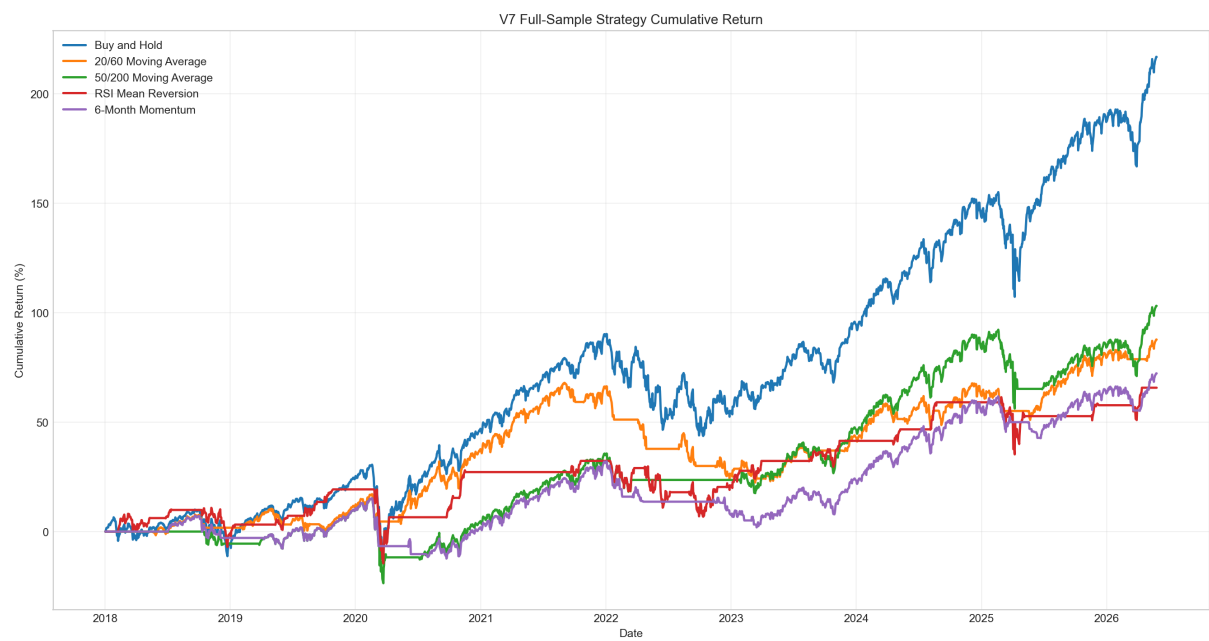
Limitations

The project only uses historical data and does not forecast future returns. It does not include slippage, taxes, or liquidity constraints. A single-asset test cannot represent every market environment.

Future Improvements

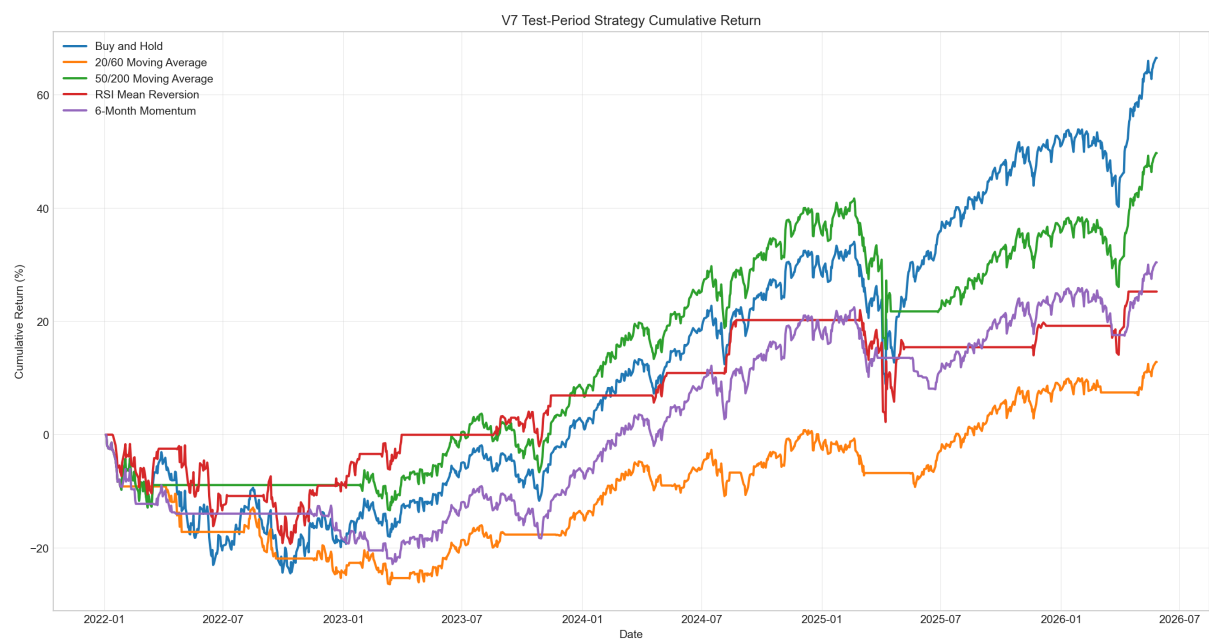
Potential extensions include multi-asset portfolios, risk parity, parameter optimization, walk-forward validation, Monte Carlo simulation, macro factors, and machine learning models.

Charts



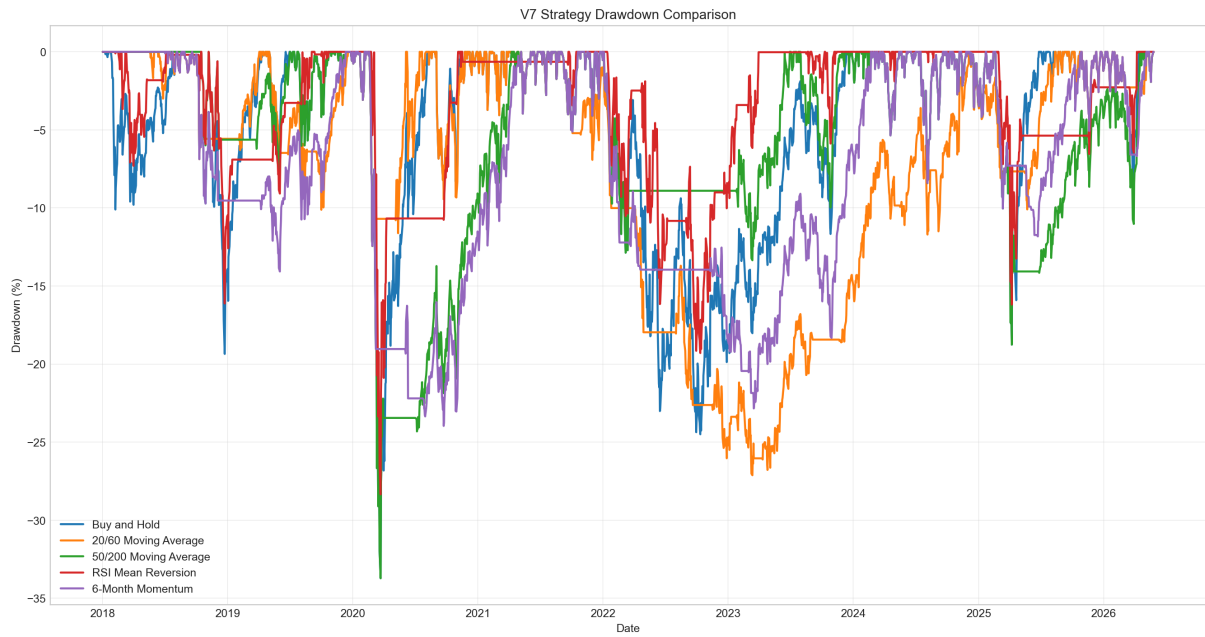
v7_strategy_comparison_full.png

Full-sample cumulative return comparison.



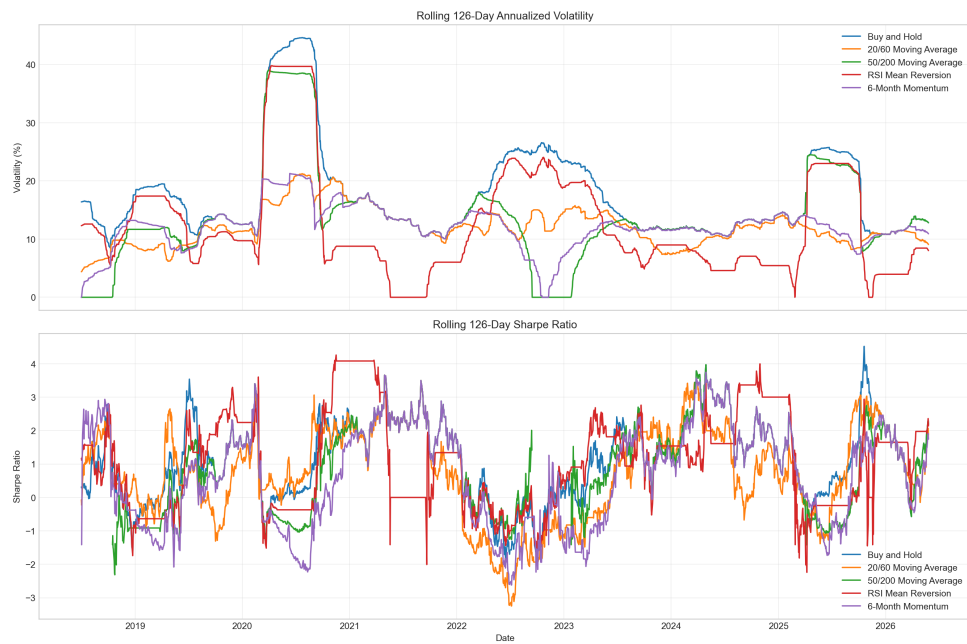
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Testing-period cumulative return comparison.



v7_drawdown_comparison.png

Drawdown comparison across strategies.



v7_rolling_risk.png

Rolling volatility and rolling Sharpe Ratio.

Disclaimer

This project is for educational and research purposes only and does not constitute investment advice.